

ANSHU

Thurana, Haryana, India - 125033

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Research Interests

- Efficient optimization strategies, Deep learning, Scientific Machine Learning(SciML), PDEs

Education

- Master of Science Mathematics | NIT Kurukshetra (2023-25) | Gold-Medalist with a 9.5422 CGPA
- Bachelor of Science Mathematics | University of Delhi (2020-2023) | First Division with a 9.324 CGPA

Master's Research

- Investigated the effectiveness of Particle Swarm Optimization for stabilizing the training of Physics-Informed Neural Network applied to elliptic, parabolic, and hyperbolic partial differential equations. Under the supervision of Dr. Raghendra Pratap Singh

Articles

- *A Hybrid Particle Swarm and Gradient-Based Optimization Technique for Training Physics-Informed Neural Networks to Solve Partial Differential Equations* (Under review)
- *A numerical scheme utilizing exponential fitting to solve a singularly perturbed differential-difference equation featuring a boundary layer phenomenon* (Under review)

Worked on

- Applied the master's research framework on 3D & 5D Burgers' and Navier-Stokes equations.
- Developed a modular Python package for few Numerical methods in masters coursework

Academic Achievements

- Top 5.7322% in GATE Mathematics 2025 (AIR 755)
- Top 5% NPTEL Number Theory Course
- Secured Category 3 in CSIR NET December 2024
- Secured Category 3 in CSIR NET June 2024
- Qualified IIT JAM 2023 in mathematics
- Qualified CUET PG 2023 in mathematics

Workshops

- Two day workshop on, Surrogates and Dimension Reduction in Scientific Machine Learning by University of Manchester, UK
- Five day workshop on, Recent Advances in Scientific Computing by Department of Mathematics NIT Kurukshetra
- One day workshop on, Diffusion Driven Instability - Non-Linear PDE by Anant The Mathematical Society NIT Kurukshetra
- One day workshop on, Vedic Mathematics - Vedic Intelligence by Anant The Mathematical Society NIT Kurukshetra

Roles & Responsibilities

- Education Outreachteam Head of Anant The Mathematical Society, NIT Kurukshetra May 2024 - May 2025.
- *Team Head of Techspardha'25 ,Annual Tech Fest of NIT Kurukshetra March 2025.*
- *Math Tutition Teacher* at Hansi April 2020 - June 2021.

Hobbies

- Exploring the global cultures
- Enthusiastic about international cuisines
- Watching the sky, searching for the limit of all limits

Language

- C1 IELTS in English
- C2 level in Hindi
- Basic Punjabi
- Native Haryanvi speaker

Professional References

Dr. Raghvendra Pratap Singh

Assistant Professor

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Declaration

I do hereby declare that all the details furnished above are true to the best of my knowledge and belief.

Place: Thurana, Haryana (India)

Anshu